

Michael Dickinson

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EDUCATION

University of British Columbia

Graduation: Apr 2027

Computer Science, Minor in Mathematics (BA) – **GPA 4.3/4.33**

Vancouver, BC

- **Teaching Assistant** for CPSC 110 (Systematic Programming Design)
- Relevant Coursework: Data Structures and Algorithms, Honors Linear Algebra, Object-Oriented Programming, Operating Systems, Computational Methods, Data Science, Mathematical Proofs

TECHNICAL SKILLS

Languages: TypeScript/JavaScript, Java, Python, C/C++, HTML/CSS, SQL

Frameworks: React.js, React Native, Flask, Express, MongoDB, PostgreSQL, Jest, PyTorch, Pandas

Tools: Git, Docker, Kubernetes, AWS, GCP, GitHub Actions CI/CD, Linux

TECHNICAL EXPERIENCE

UBC Uncrewed Aircraft Systems

Sep 2024 – Present

Software Engineer

Vancouver, BC

- Led Ground Communications team to **National 1st Place Win** at SUAS 2025 - integrated with 30+ members
- Created integration testing suite with Pytest and **Docker Compose**, eliminating **11 failing flight situations** and leading to first successful test flight – deployed to **GitHub Actions** CI/CD to support reliability
- Deployed low-latency **WebRTC** aerial image streaming, enabling **real-time drone feeds** during autonomous flight – deployed to **AWS** with **EC2** signaling server and **S3** persistence
- Developed primary communication link between ground station and drone via **WebSocket** handler with **Redis** caching for live telemetry retrieval during autonomous flight

DevSwarm

Jun 2025 – Sep 2025

Software Engineering Intern

Portland, OR

- Sustained **16+ concurrent tasks** without UI stalls by implementing a **Node.js worker-pool executor** with back-pressure, retries, and cancellation to isolate CPU-intensive operations from Electron's main thread.
- Delivered cross-platform compatible **Git flows** (branching, PR preview, conflict wizard) in **Electron/React**, achieving **50% adoption** among beta users and driving 15% more commits completed in-app

Twenty Ideas

Apr 2023 – Jul 2024

Software Engineer

Eugene, OR

- Reduced QA turnaround times to < 20 minutes/build, enabling **nightly releases** via Github Actions **CI/CD** pipeline, integrating with AWS for infrastructure deployments on **AWS Fargate + ECS**
- Integrated **AWS CDK IaC** workflows into deployment pipeline, allowing for infrastructure deployments in under 10 minutes, including plan/apply stages for multiple environments and audit trail + rollback capability
- Architected **REST API** consolidating **4,000+** lines of legacy code into reusable service layers with **gRPC** for inter-service communication and **Cloudwatch** monitoring for end-to-end observability of **5,000+** daily requests

SheerID

Apr 2022 – Sep 2022

Data Science Intern

Portland, OR

- Generated over **1000** conversions for clients such as **Google** and **AT&T** leveraging **Selenium web scraping** and **AWS Lambda** ETL pipeline to **locate over 10,000 students** and provide them with custom promotions
- Utilized **SQLAlchemy + Python Pandas** to scrub **Postgres database**, removing 300+ redundant user records

PROJECTS

DubHacks 2025 – E2E Test Automation Platform | *Python, FastAPI, Kubernetes, Playwright, OpenAI*

- Built end-to-end testing platform converting natural language to Playwright scripts via OpenAI Agents and **MCP**
- Built **FastAPI** backend with **Kubernetes** isolated test execution and CPU/RAM limits to prevent resource leaks
- Integrated live result tracking and flaky test detection with **PostgreSQL** and **Grafana** dashboard for observability

BERT + Vision Transformer – Multimodal Suicide Prevention Research | *Pytorch, Numpy, OpenCV*

- Engineered novel cross-attention architecture extending BERT for multimodal analysis of social media content (text + images), achieving 96% accuracy in suicidal ideation detection across 5,000 posts
- Awarded **American Psychological Association Achievement Award** for research on suicide intervention